

FIG. 1 — FRONT VIEW Intersection Core™

Supercapacitor-assisted backup power for traffic intersections

- **4,800 Wh LiFePO4 battery backup** for signal cabinets and communications
- **62.1V / 130F supercapacitor buffer** for seamless power transitions from grid to backup power.
- **<20 ms UPS transfer** — controller and network equipment stay online
- **Insulated internal battery enclosure -** extend battery life against harsh weather conditions with outdoor rated IP connector interface.
- **Remote monitoring via Wi-Fi** for city fleet visibility of units
- **Aluminum outer case (stainless steel option) -** conform to FDOT standard* and protect against environmental hazards

4,800 Wh BATTERY ENERGY	62.1 V SUPERCAP RATED	<0.03 W/(m·K) INSULATION (k)	<20 ms UPS TRANSFER	35 AMP CHARGER	~210 LBS EST. SYSTEM WEIGHT

INTERSECTION APPLICATIONS

Application	Loads Supported
Signal cabinet backup	Controller, LED heads, conflict monitor, auxiliaries
Communications continuity	Router, radio, fiber modem, telemetry gateway
Temporary construction signal	Portable cabinet, work-zone signals, monitoring
Emergency / storm recovery	Critical intersection loads and network devices

KEY SPECIFICATIONS

Energy capacity	**4,800 Wh** LiFePO4 gross battery energy	**Battery**	**48V 100Ah LiFePO4** BMS protected
DC Output	**12V/24V/48V** DC voltages available	**Supercapacitor**	**62.1V / 130F**
Supercap buffer	**Contact us** Usable energy	**AC input**	**90–140 VAC** Wall · generator · field power source
Inverter / charger	**48 V / 3,000 VA** Integrated inverter/charger	**Recharge**	**Contact us** 35A integrated DC Charger
AC output	**3,000 VA** 120V 60Hz critical cabinet output	**Monitoring**	**Touchscreen controller / remote monitoring** Wi-Fi · Bluetooth · Ethernet
UPS transfer	**<20 ms** Controller/network ride-through	**Internal Power Wire**	**4/0 AWG** Assist with handling surge in power

* In progress

Given ambient weather conditions of 75 °F ± 10 °F & 50% ± 10% RH Values may vary outside of these conditions

ADDITIONAL SPECIFICATIONS

Surge capacity	**Contact us** Inverter peak	**Battery weight**	**~90 lbs**
Supercap pulse	**Contact us**	**Enclosure**	**0.125" Aluminum** Stainless steel enclosure option also available
Supercap current	**Contact us** Pulse current rating	**Cooling**	**2× 120mm fans** With IP66 Sealing*
Nominal voltage	**48 V**	**Charge voltage**	**54.75 V**
Float voltage	**52.0 V**	**Operating voltage range**	**42.0 - 54.75 V**
Low-voltage cut-off	**42.0 V**	**Internal Insulation**	**<0.03 W/(m·K)** High insulation rating
Efficiency	**Contact us**	**Communication**	**Inverter comm bus + BMS-CAN + Ethernet**
Max battery discharge rate	**Contact us**	**Protection**	**Multiple 125A UL-rated fuse + breaker**

Operating temperature range, enclosure NEMA/IP rating, runtime at load, mounting and dimensions: contact us.

CERTIFICATIONS & COMPLIANCE

Standard / Area	Scope
UL 1778* / UL 9540*	Uninterruptible power supply / energy storage system
UN 38.3* / UL 1973*	LiFePO4 battery transport and stationary battery safety
UL 508A* / NEMA TS2*	Industrial control panel / traffic cabinet equipment
FCC Part 15*	Emissions / communications

WARRANTY & SUPPORT

Coverage	Details
Warranty	Contact us for warranty details.

* In progress